## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (http://bmjopen.bmj.com/site/about/resources/checklist.pdf) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Assessment of patient-reported outcomes after polytrauma – instruments and methods: a systematic review |
| **AUTHORS** | Ritschel, Michaela; Kuske, Silke; Gnass, Irmela; Andrich, Silke; Moschinski, Kai; Borgmann, Sandra; Herrmann-Frank, Annegret; Metzendorf, Maria-Inti; Wittgens, Charlotte; Flohé, Sascha; Sturm, Johannes; Windolf, Joachim; Icks, Andrea |

## VERSION 1 – REVIEW

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| **REVIEWER** | Fontoura Solla, Davi<br>University of São Paulo Department of Neurosciences and Behaviour Sciences |
| **REVIEW RETURNED** | 18-Mar-2021 |

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| **GENERAL COMMENTS** | This is a great systematic review on an important and undervalued matter. These are my minor concerns:<br>1. Please include in the manuscript the rationale for the inclusion of papers from 2005, which is already present in the protocol. I believe this is an important information for the readers and should be in the main manuscript.<br>2. Although briefly cited in the limitations section, the language restriction should be further emphasized. That's a major limitation of this systematic review since low and middle income countries, where most of the trauma burden lies, are probably publishing local results in local languages. Replicating the PubMed search strategy only, at least >30 papers primarily classified as Spanish, Portuguese, Chinese or Turkish would be triaged. I'm not suggesting that these should be included, but just further emphasized.<br>3. Was a publication time trend identified? |

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| **REVIEWER** | Jarman, Heather<br>St George's University Hospitals NHS Foundation Trust, Emergency Department Clinical Research Unit |
| **REVIEW RETURNED** | 19-May-2021 |

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| **GENERAL COMMENTS** | Thank you for the opportunity to review this manuscript.<br>It is a systematic review of assessment of patient reported outcomes (PRO) after polytrauma. Overall the manuscript very well written and clear to read. The methods are clearly described and the structure of the paper is easy to follow. In my opinion there are some minor recommendations that would improve this work. |

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| | Title: The review is of the use of PRO assessment instruments in this patient group rather than the outcomes of the assessment of PRO. It would aid clarity to add the work “instruments” to the title.<br>Introduction: the final paragraph gives a single sentence on “participation” but the aim 1 refers to social participation. These concepts are different so please be consistent in the text.<br>Methods: this is a well written section and I note the previous publication. I notice that the inclusion criteria did not include “major trauma” which is very common in UK and North America paper, do you think this was a factor in the lack of papers from these areas? No suggestion that this impacts on the paper you have written but may limit the applicability to those countries.<br>Under the heading ‘critical appraisal’ please add the types of papers that were covered by each of the three standardised checklists. It will assist the reader with understanding in the tables for the different types of studies.<br>Results: under the heading ‘critical appraisal results’ please could you give examples of some of the confounders.<br>Discussion: I found the discussion question brief in relation to the number of papers found.<br>Under the heading ‘instruments for the assessment of HRQoL…’ it would aid the paper to provide some description of the measures within the HRQoL tools found. Are they used in different major trauma groups or were reasons identified in the original papers for selection of one version of the short-form questionnaire over another? Currently this paragraph is a ‘count’ of the number papers and could be expanded. This is included somewhat in the tables but a short descriptive section would help with the applicability to practice. If there is no pattern, then please state that.<br>Under the heading ‘application of instruments’ please add a reference the statement in the second statement on postal questionnaires being quicker. You have only provided a critique on postal collection in the text, what about the similar biases with in the other methods of collection? |

**VERSION 1 – AUTHOR RESPONSE**

Referee: 1

Paragraph 1

This is a great systematic review on an important and undervalued matter. These are my minor concerns: Please include in the manuscript the rationale for the inclusion of papers from 2005, which is already present in the protocol. I believe this is an important information for the readers and should be in the main manuscript.

Thank you for your appreciation of our systematic review! Thank you also for this important feedback. We added the rationale in the section ‘Data sources and search strategy’ on page 5 (Main Document_marked copy): “The search period was chosen because since 2005 interest in HRQoL in research and its implication in practice has been growing.[28] As one of our main aims was to collect instruments measuring HRQoL, we decided to include publications beginning with 2005. Our approach is supported by the emerging importance of PROs in clinical research and their relevance to the future as a complement to

clinical data.[29] Furthermore, the search period was selected to capture the most recent developments in research on PROs."

Paragraph 2

Although briefly cited in the limitations section, the language restriction should be further emphasized. That's a major limitation of this systematic review since low and middle income countries, where most of the trauma burden lies, are probably publishing local results in local languages. Replicating the PubMed search strategy only, at least >30 papers primarily classified as Spanish, Portuguese, Chinese or Turkish would be triaged. I'm not suggesting that these should be included, but just further emphasized.

We further emphasized the language restriction in the limitation section and hope to have addressed your comment sufficiently. Please see page 37 (Main Document_marked copy): "Our study only included English and German language literature and thus does not provide insight into published results in other languages. Therefore, it cannot be ruled out that relevant publications from low and middle income countries containing local results in local languages were not included. However, we assume that scientific articles are predominantly published in English; thus, the results presented here should give a comprehensive impression related to the research questions of this systematic review."

Paragraph 3

Was a publication time trend identified?
We shortly referred to this in our Results in the section ‚Characteristics of included publications' – please see page 8 (Main Document_marked copy):
More than half of the publications appeared between 2005 and 2009; the number of annual publications seems to be decreasing since 2015.

Referee: 2

Comments to the Authors

Paragraph 1

Thank you for the opportunity to review this manuscript.
It is a systematic review of assessment of patient reported outcomes (PRO) after polytrauma. Overall the manuscript very well written and clear to read. The methods are clearly described and the structure of the paper is easy to follow. In my opinion there are some minor recommendations that would improve this work.
Title: The review is of the use of PRO assessment instruments in this patient group rather than the outcomes of the assessment of PRO. It would aid clarity to add the work "instruments" to the title.

Thank you for your appreciation of our systematic review. We have added an addition to the title regarding the instruments. The new title is: "Asessment of patient-reported outcomes after polytrauma – instruments and methods: a systematic review "

Paragraph 2

Introduction: the final paragraph gives a single sentence on "participation" but the aim 1 refers to social participation. These concepts are different so please be consistent in the text.

Thank you for your helpful comment here. We have clarified this accordingly – please see page 4 (Main Document_marked copy): We have deleted the following sentence "Participation is a health component according to the International Classification of Functioning, Disability and Health (ICF)." The remaining sentence is now states as follows: "In recent years, the concept of social participation has

gained more attention in the political context [17] and was therefore included in the review as an outcome of interest, along with HRQoL and ADL."

Paragraph 3

Methods: this is a well written section and I note the previous publication. I notice that the inclusion criteria did not include "major trauma" which is very common in UK and North America paper, do you think this was a factor in the lack of papers from these areas? No suggestion that this impacts on the paper you have written but may limit the applicability to those countries.

This is a very important point. We made a conscious decision not to include the term "major trauma" in our search strategy with the aim of capturing publications reporting severe injuries with an ISS >15. At the same time, we wanted to avoid finding more studies reporting single injuries with study populations and an ISS less than 16, which would have increased the number of hits without adding content value. "Major trauma" seemed to be increasingly mentioned in connection with single and psychological trauma - neither of which met the inclusion criteria of our systematic review and would have led to an increase in the number of hits. In addition, we screened systematic reviews and meta-analyses to identify further appropriate studies and would have included publications reporting major trauma if they fulfilled the inclusion criteria (ISS>15, exclusion of single trauma). In summary, we do not believe that this fact is a reason for the absence of papers from the UK and North America. On the other hand, we cannot completely exclude this circumstance. As a consequence we added the following phrase in the limitation section – please see page 37 (Main Document_marked copy):
"We did not use the term 'major trauma'. Therefore, we cannot rule out that relevant publications reporting major trauma were not included. However, major trauma seemed to be increasingly mentioned in connection with single and psychological trauma - neither of which met the inclusion criteria of our systematic review. We screened systematic reviews and meta-analyses to identify further appropriate studies and would have included publications reporting major trauma if they fulfilled the inclusion criteria of this systematic review. Still, we believe that the non-use of the term 'major trauma' has no impact on the results of this systematic review and the answers to the research questions."

Paragraph 4

Under the heading 'critical appraisal' please add the types of papers that were covered by each of the three standardised checklists. It will assist the reader with understanding in the tables for the different types of studies.
Results: under the heading 'critical appraisal results' please could you give examples of some of the confounders.

We added these informations accordingliy – please see in the Results the section ‚Critical appraisal results' on page 33 (Main Document_marked copy): "The methodological quality and the risk of bias was appraised using the NICE-checklist for quantitative intervention studies,[32] [40,67] the NICE-checklist for quantitative studies reporting correlations and associations,[33] [1, 7, 11, 36-39, 42-66, 68-82, 85, 86] the SIGN-checklist for diagnostic studies,[34] [41, 83] and the Mixed Method Appraisal Tool.[35] [84]"

"Some methodological weaknesses were identified regarding the selection of the study population, consideration of confounders (e.g. age, sex), data analysis and presentation of results, internal validity and generalisability of study results."

Paragraph 5

Discussion: I found the discussion question brief in relation to the number of papers found.
Under the heading 'instruments for the assessment of HRQoL…' it would aid the paper to provide some description of the measures within the HRQoL tools found. Are they used in different major trauma groups or were reasons identified in the original papers for selection of one version of the short-form questionnaire

over another? Currently this paragraph is a ‘count’ of the number papers and could be expanded. This is included somewhat in the tables but a short descriptive section would help with the applicability to practice. If there is no pattern, then please state that.

We agree with the reviewer that some description of the measures within the HRQoL tools found would be interesting. Since this is not the focus of this systematic review and primarily established and widely known instruments are found, we decided not to amend this information.
No pattern could be found regarding the use of instruments in different trauma groups and no explanation was given in the publications for the decision of the SF-12 over the SF-36. We added this information in the Discussion on page 34 (Main Document_marked copy): “Considering the use of the different instruments assessing HRQoL by the different trauma groups, it becomes apparent that instruments are used without a clearly recognizable pattern or reason. Similarly, no pattern can be identified to explain the use of the SF-12 instead of the SF-36. No information was provided in this regard in the publications. However, it can be assumed that the reason for using the SF-12 is that it takes less time to administer than the SF-36.”

Paragraph 6

Under the heading ‘application of instruments’ please add a reference the statement in the second statement on postal questionnaires being quicker. You have only provided a critique on postal collection in the text, what about the similar biases with in the other methods of collection?

We added these informations accordingliy – please see in the Discussion the section ‚Application of instruments‘ on page 35 (Main Document_marked copy): “Most frequently, a postal questionnaire was used. Compared to personal interviews, questionnaires sent by mail are quicker and less expensive.[99] […] However, other methods of data collection like personal or telephone interviews also have limitations (e.g. influence of the interviewer on the person being interviewed),[99] thus all data collection methods may incorporate (different) biases.”

**VERSION 2 – REVIEW**

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| **REVIEWER** | Fontoura Solla, Davi<br>University of São Paulo Department of Neurosciences and Behaviour Sciences |
| **REVIEW RETURNED** | 28-Jul-2021 |

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| **GENERAL COMMENTS** | The corrections improved the paper. |